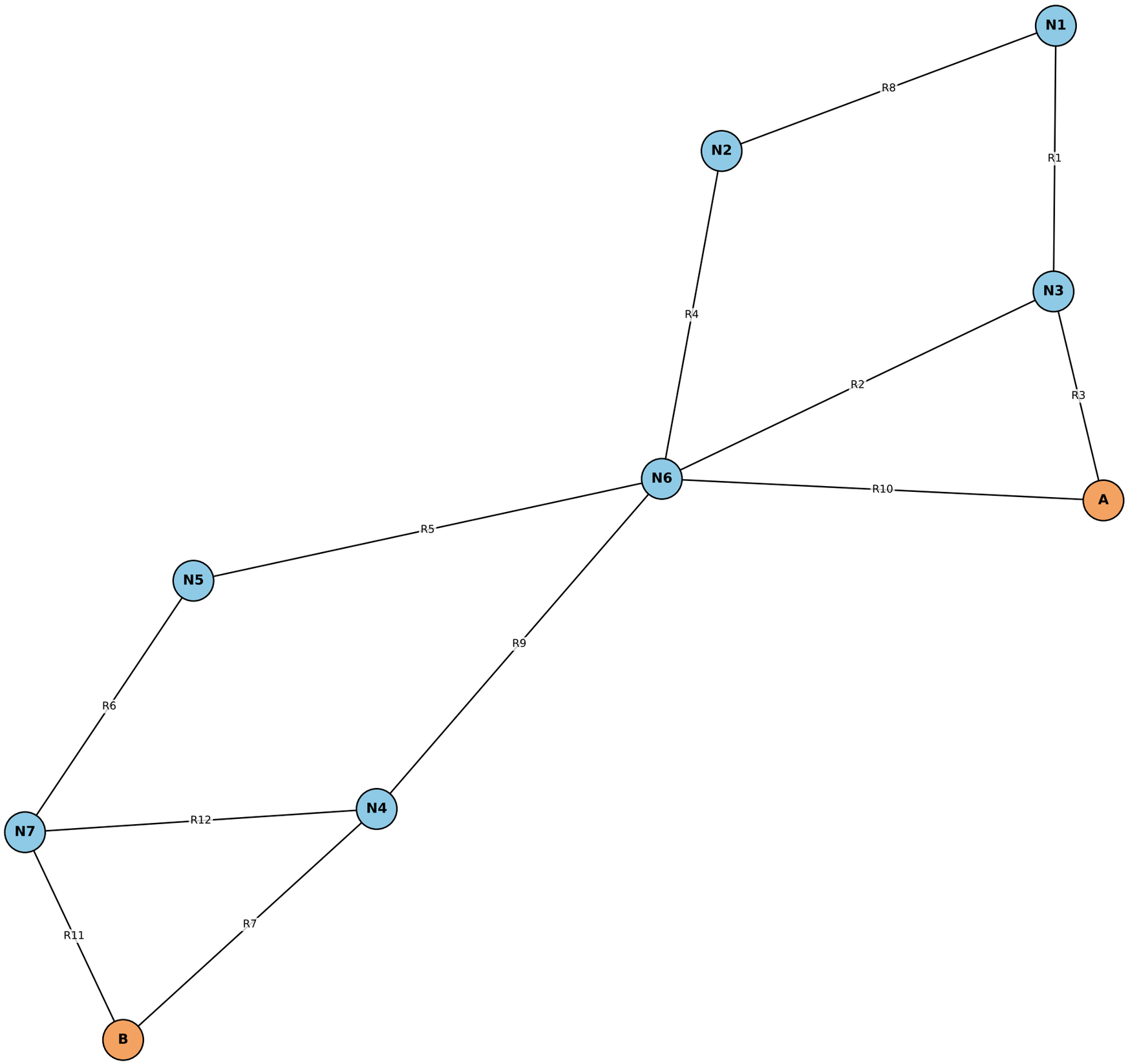
Resistor network v4 · Level 1 · task_0 · 12 resistors · 9 nodes

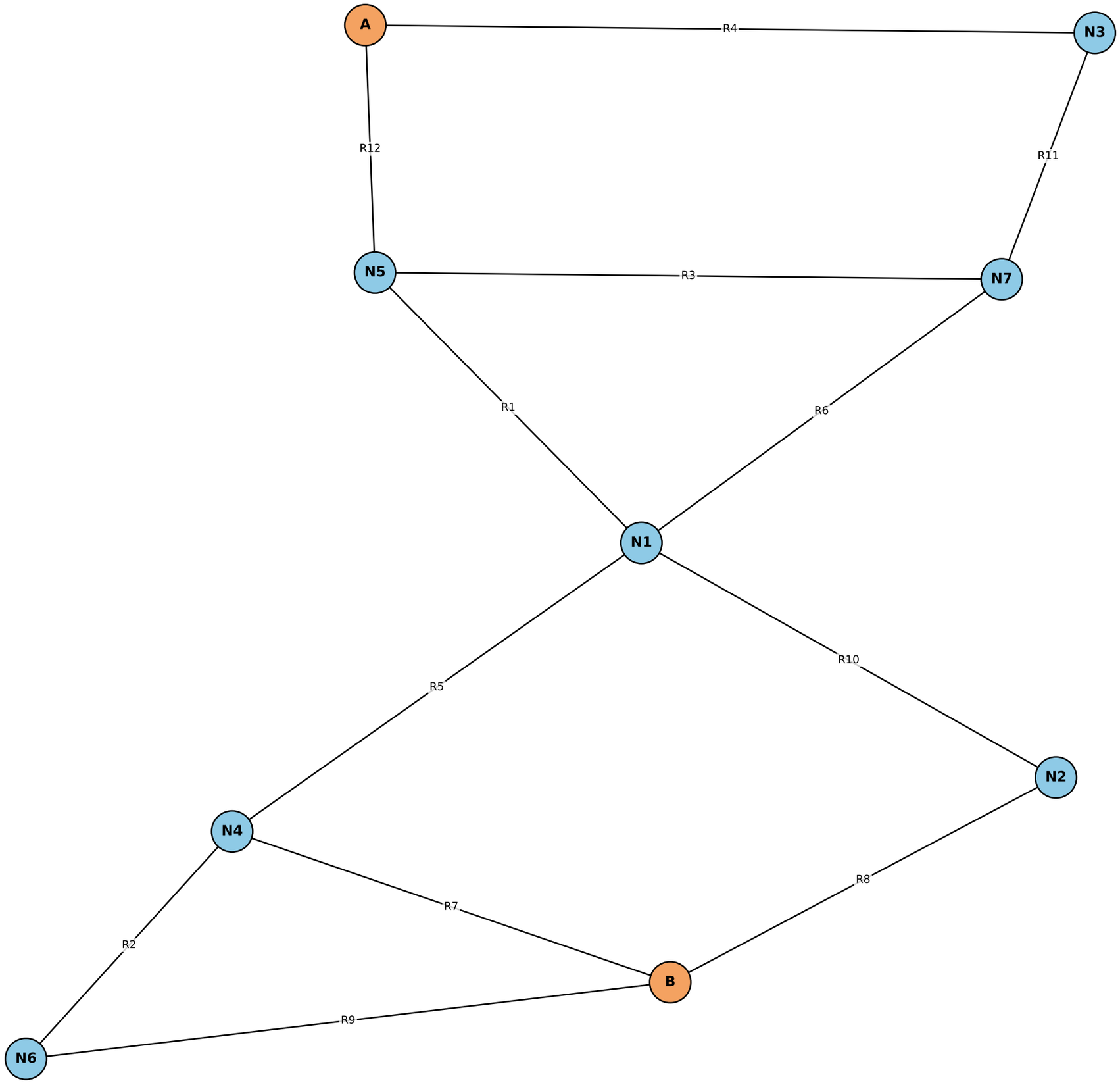
4 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N3-N1	15.000000
R2	N6-N3	119.000000
R3	A-N3	108.000000
R4	N2-N6	119.000000
R5	N6-N5	28.000000
R6	N5-N7	28.000000
R7	N4-B	71.000000
R8	N2-N1	64.000000
R9	N4-N6	14.000000
R10	N6-A	23.000000
R11	N7-B	88.000000
R12	N4-N7	79.000000

Resistor network v4 · Level 1 · task_1 · 12 resistors · 9 nodes

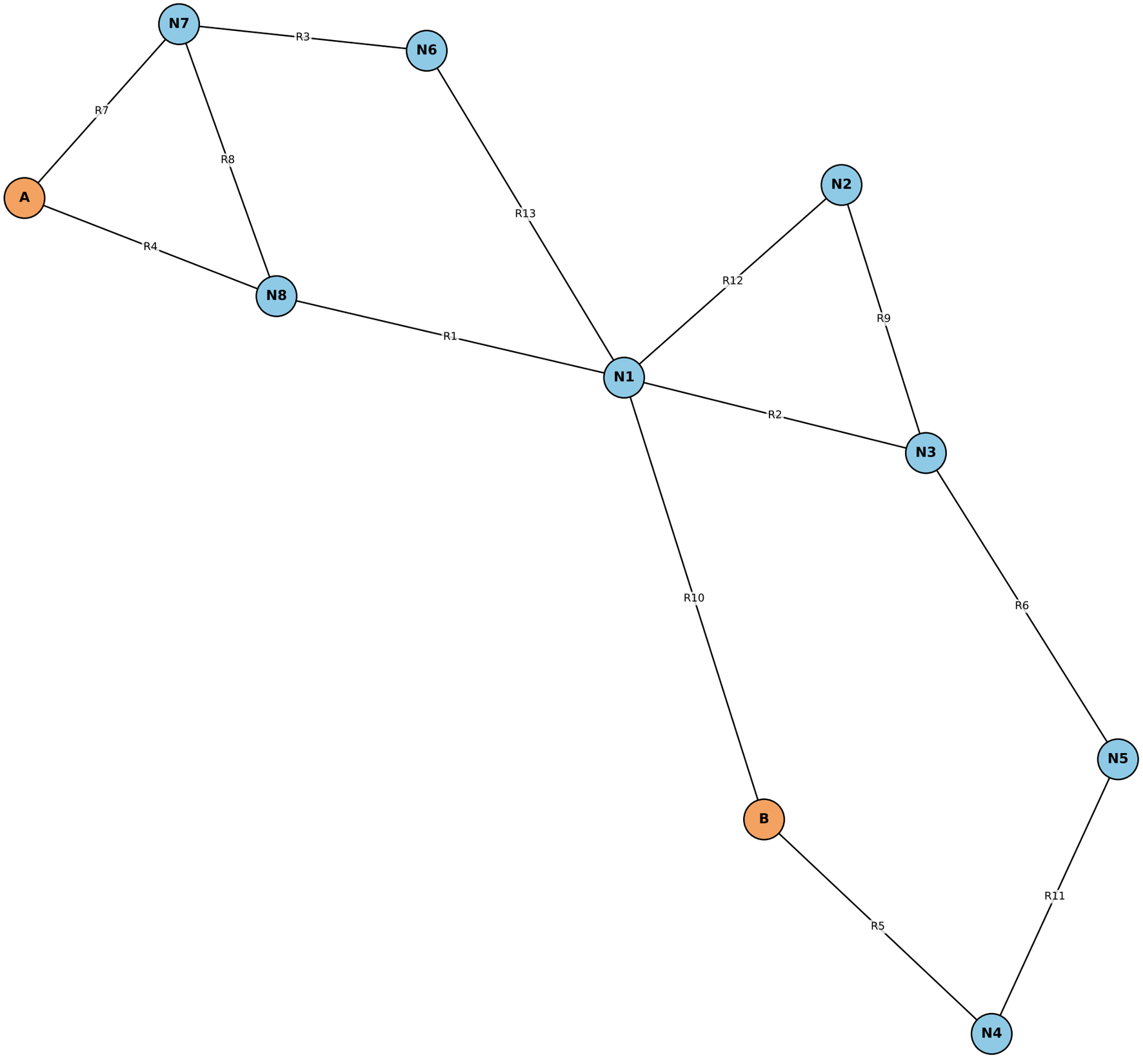
4 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N5-N1	180.000000
R2	N4-N6	58.000000
R3	N5-N7	38.000000
R4	N3-A	119.000000
R5	N1-N4	11.000000
R6	N1-N7	28.000000
R7	N4-B	42.000000
R8	N2-B	132.000000
R9	N6-B	42.000000
R10	N2-N1	17.000000
R11	N7-N3	71.000000
R12	N5-A	11.000000

Resistor network v4 · Level 1 · task_2 · 13 resistors · 10 nodes

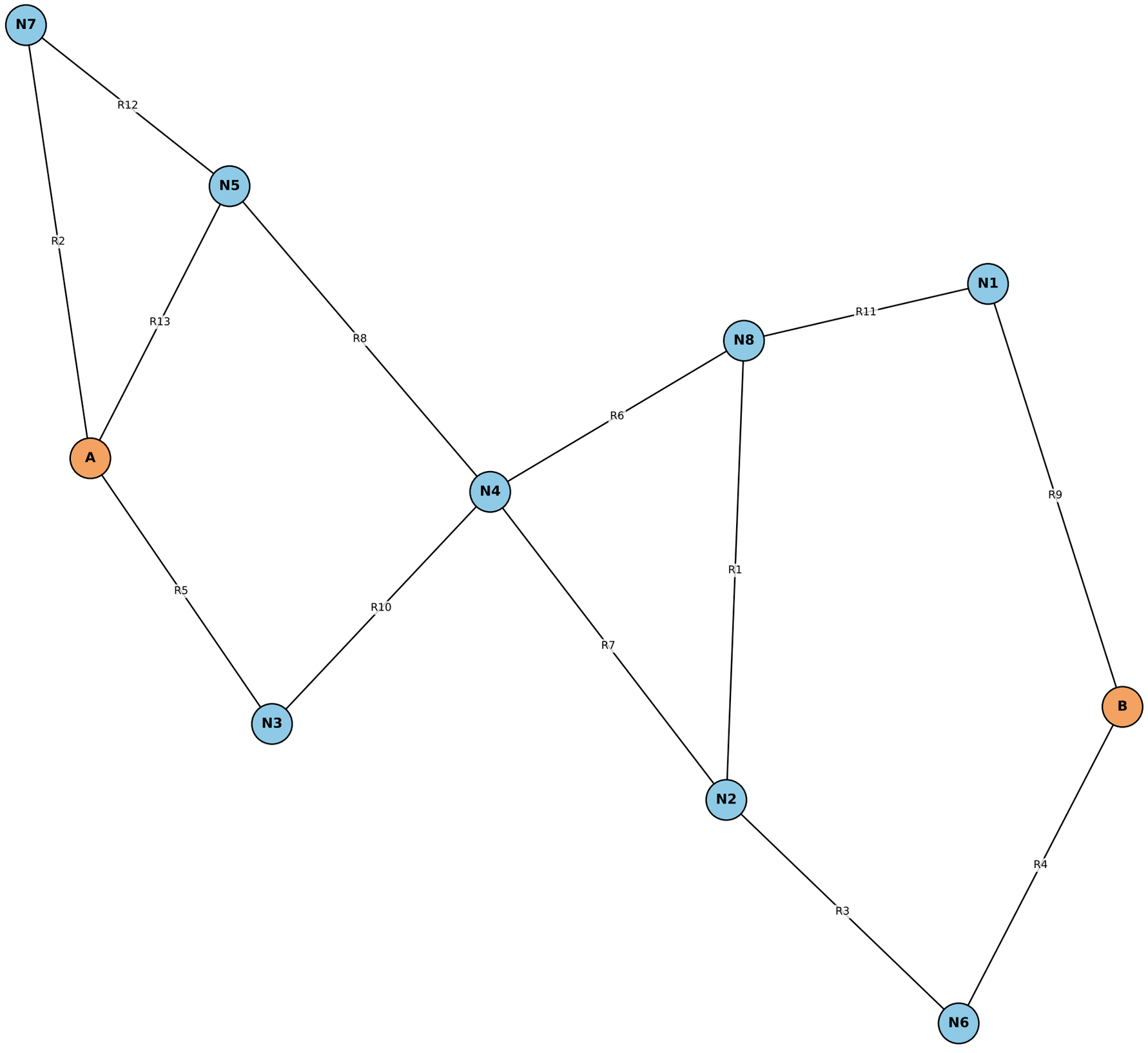
4 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N1-N8	25.000000
R2	N3-N1	19.000000
R3	N6-N7	200.000000
R4	N8-A	38.000000
R5	N4-B	25.000000
R6	N3-N5	97.000000
R7	A-N7	17.000000
R8	N8-N7	15.000000
R9	N3-N2	38.000000
R10	B-N1	79.000000
R11	N5-N4	71.000000
R12	N1-N2	200.000000
R13	N6-N1	180.000000

Resistor network v4 · Level 1 · task_3 · 13 resistors · 10 nodes

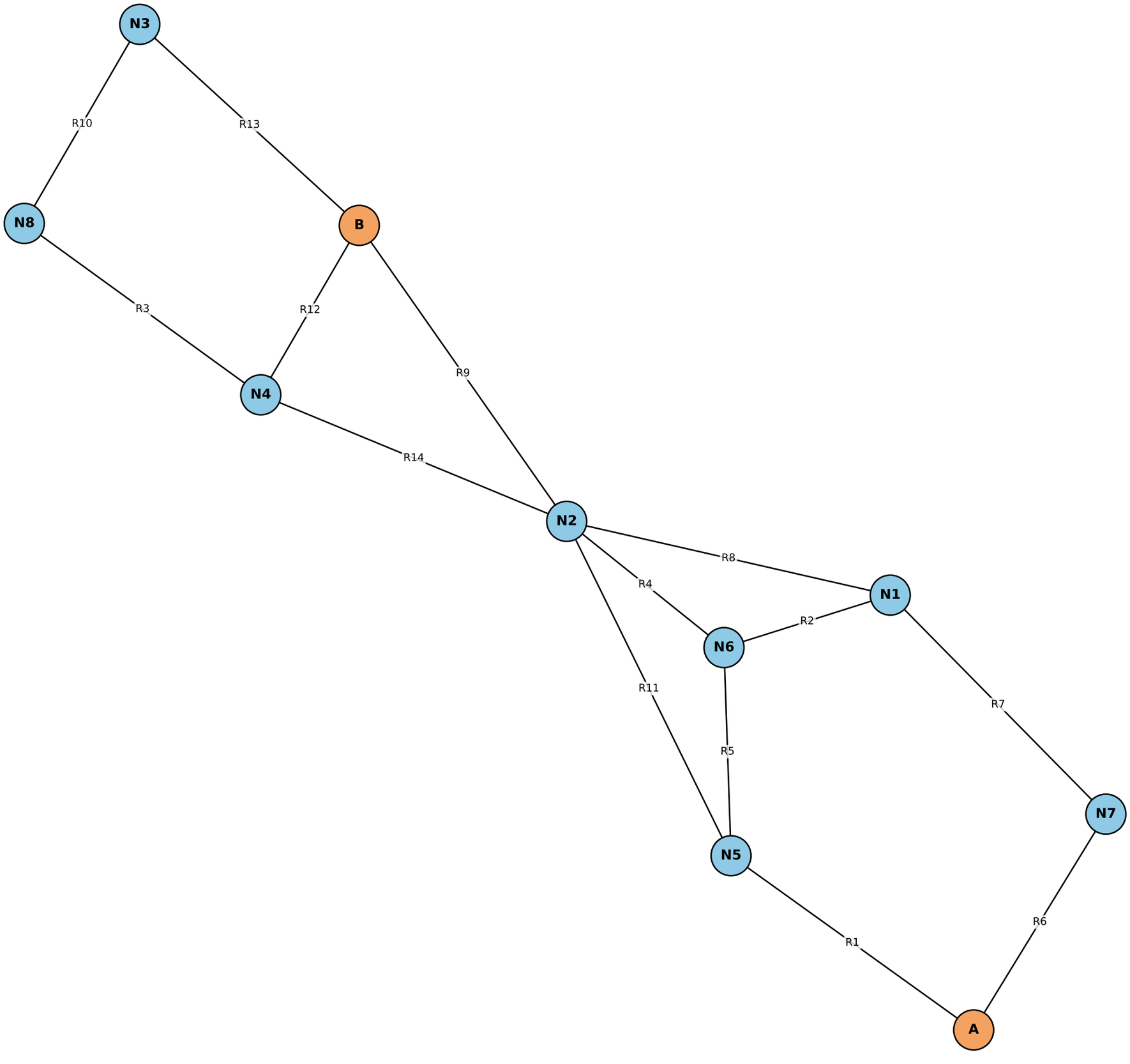
4 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N8-N2	163.000000
R2	N7-A	147.000000
R3	N6-N2	14.000000
R4	N6-B	42.000000
R5	A-N3	23.000000
R6	N8-N4	147.000000
R7	N4-N2	64.000000
R8	N5-N4	31.000000
R9	B-N1	23.000000
R10	N3-N4	21.000000
R11	N1-N8	58.000000
R12	N5-N7	79.000000
R13	N5-A	147.000000

Resistor network v4 · Level 1 · task_4 · 14 resistors · 10 nodes

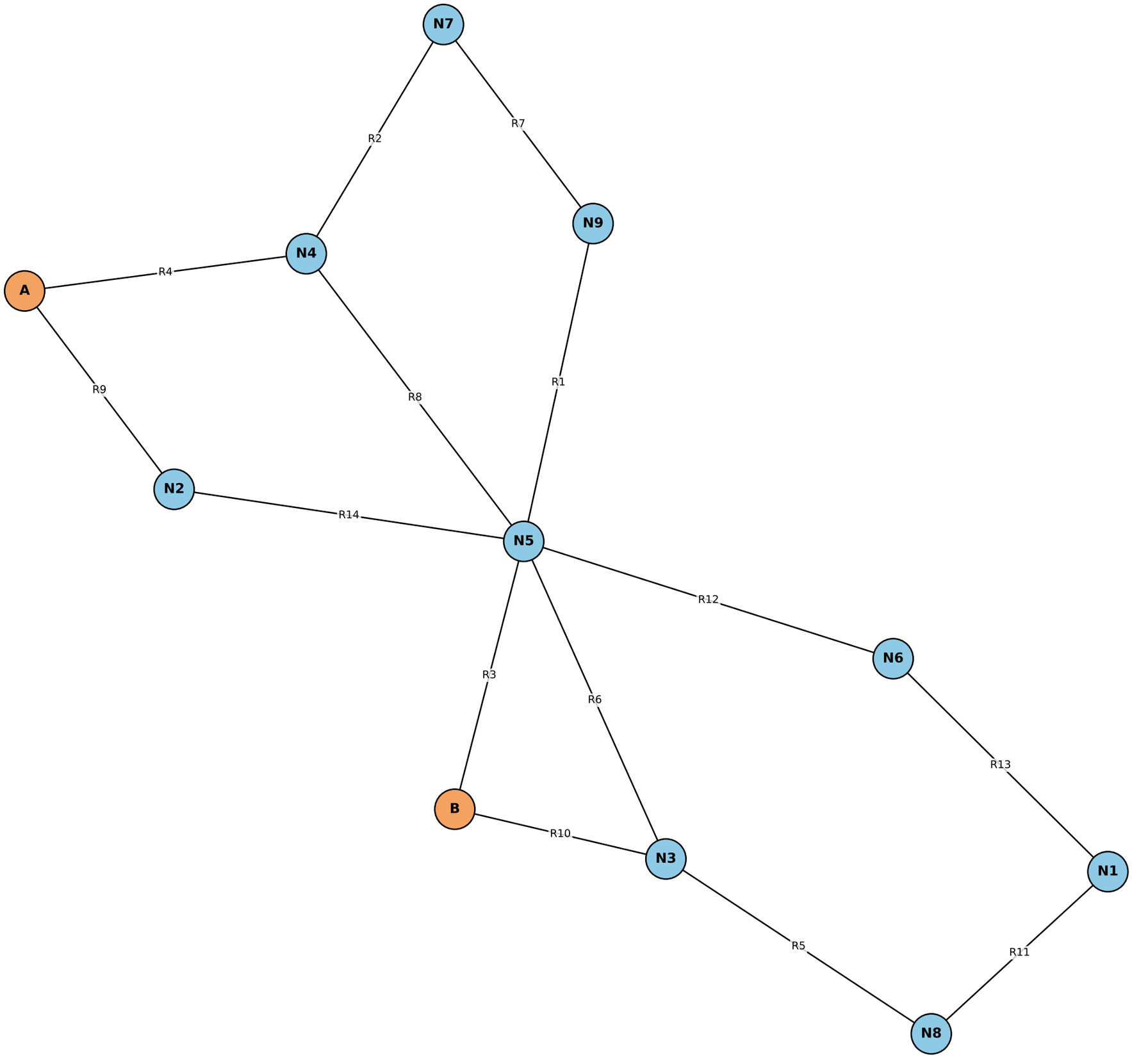
5 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	A-N5	25.000000
R2	N1-N6	25.000000
R3	N8-N4	163.000000
R4	N2-N6	58.000000
R5	N6-N5	12.000000
R6	A-N7	58.000000
R7	N7-N1	132.000000
R8	N1-N2	180.000000
R9	B-N2	25.000000
R10	N8-N3	11.000000
R11	N2-N5	11.000000
R12	B-N4	163.000000
R13	N3-B	10.000000
R14	N2-N4	31.000000

Resistor network v4 · Level 1 · task_5 · 14 resistors · 11 nodes

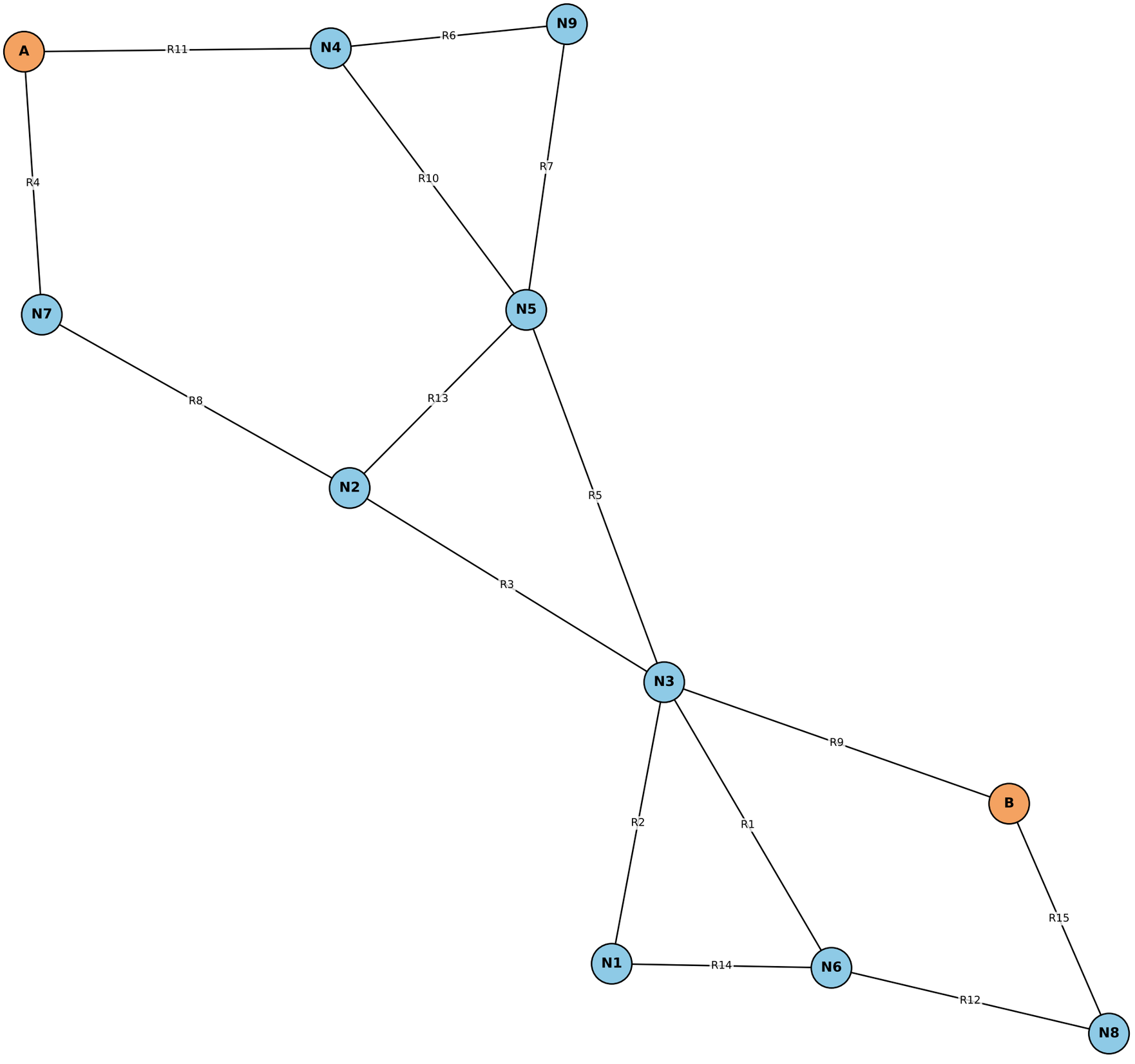
4 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N9-N5	64.000000
R2	N4-N7	200.000000
R3	B-N5	21.000000
R4	A-N4	42.000000
R5	N8-N3	12.000000
R6	N3-N5	38.000000
R7	N9-N7	15.000000
R8	N5-N4	38.000000
R9	N2-A	163.000000
R10	N3-B	21.000000
R11	N8-N1	17.000000
R12	N5-N6	88.000000
R13	N6-N1	42.000000
R14	N5-N2	38.000000

Resistor network v4 · Level 1 · task_6 · 15 resistors · 11 nodes

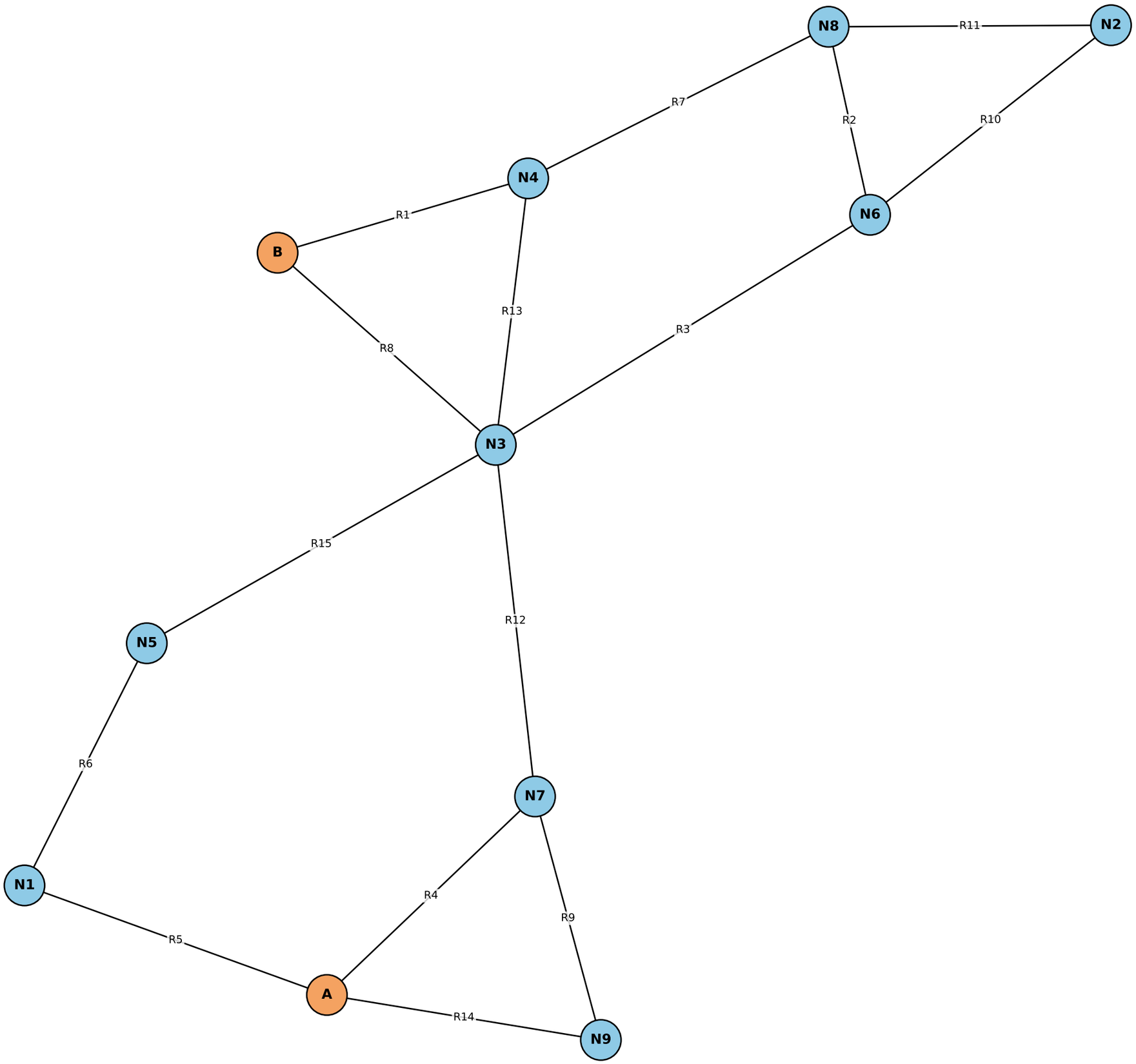
5 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N6-N3	14.000000
R2	N1-N3	47.000000
R3	N3-N2	12.000000
R4	N7-A	71.000000
R5	N5-N3	15.000000
R6	N4-N9	35.000000
R7	N9-N5	163.000000
R8	N2-N7	42.000000
R9	N3-B	42.000000
R10	N4-N5	12.000000
R11	N4-A	163.000000
R12	N8-N6	132.000000
R13	N5-N2	14.000000
R14	N6-N1	10.000000
R15	N8-B	58.000000

Resistor network v4 · Level 1 · task_7 · 15 resistors · 11 nodes

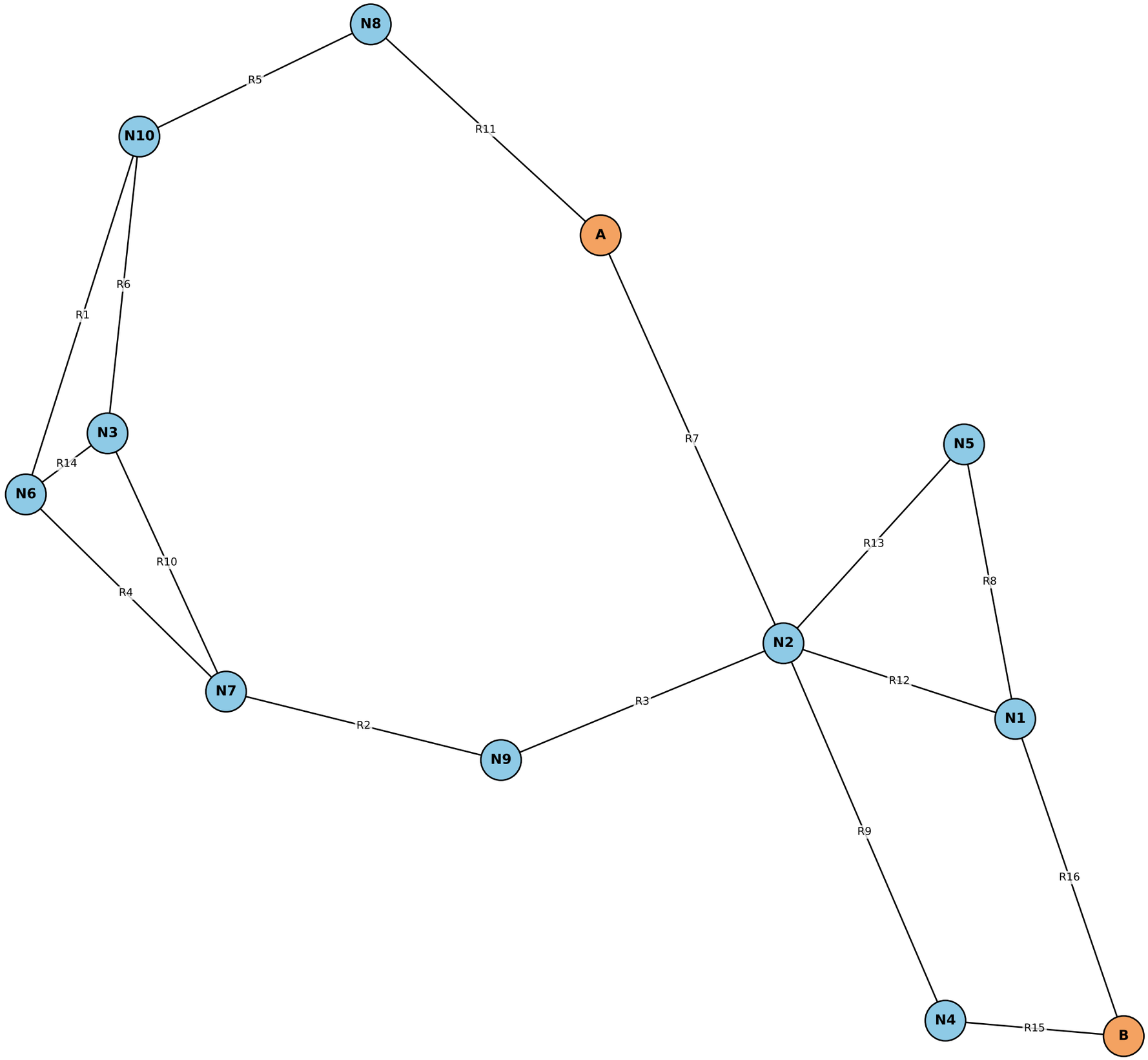
5 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	B-N4	14.000000
R2	N8-N6	23.000000
R3	N3-N6	14.000000
R4	N7-A	15.000000
R5	N1-A	64.000000
R6	N5-N1	12.000000
R7	N4-N8	28.000000
R8	N3-B	47.000000
R9	N7-N9	17.000000
R10	N2-N6	31.000000
R11	N8-N2	97.000000
R12	N7-N3	38.000000
R13	N3-N4	12.000000
R14	A-N9	21.000000
R15	N3-N5	132.000000

Resistor network v4 · Level 1 · task_8 · 16 resistors · 12 nodes

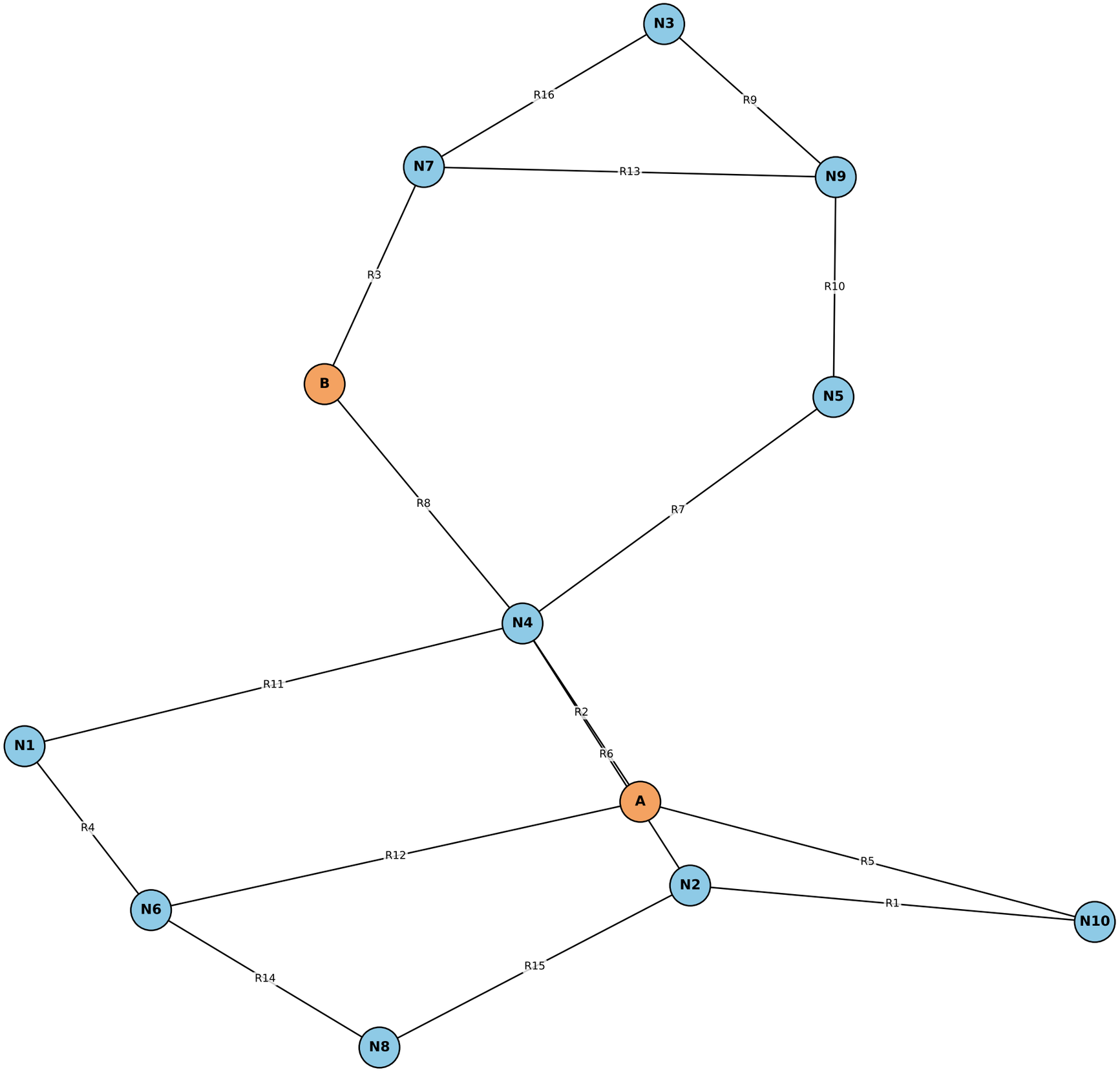
5 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N10-N6	35.000000
R2	N7-N9	180.000000
R3	N9-N2	108.000000
R4	N6-N7	42.000000
R5	N8-N10	15.000000
R6	N10-N3	35.000000
R7	N2-A	14.000000
R8	N5-N1	17.000000
R9	N4-N2	15.000000
R10	N3-N7	64.000000
R11	N8-A	14.000000
R12	N1-N2	180.000000
R13	N5-N2	132.000000
R14	N6-N3	21.000000
R15	B-N4	71.000000
R16	N1-B	200.000000

Resistor network v4 · Level 1 · task_9 · 16 resistors · 12 nodes

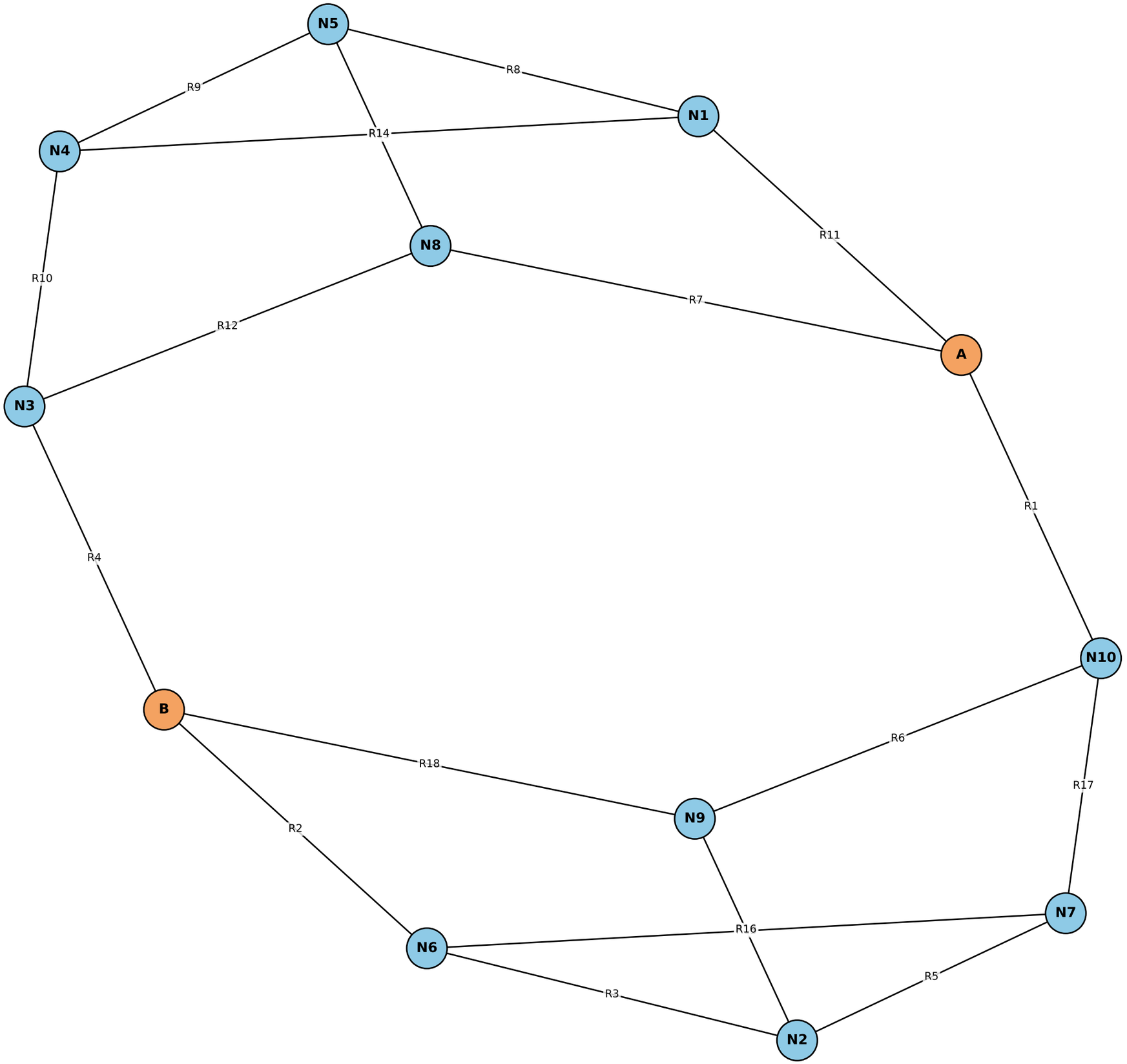
5 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N2-N10	10.000000
R2	A-N4	12.000000
R3	N7-B	147.000000
R4	N6-N1	11.000000
R5	N10-A	14.000000
R6	N2-N4	147.000000
R7	N4-N5	35.000000
R8	N4-B	163.000000
R9	N9-N3	52.000000
R10	N5-N9	132.000000
R11	N4-N1	108.000000
R12	N6-A	14.000000
R13	N9-N7	47.000000
R14	N6-N8	25.000000
R15	N8-N2	88.000000
R16	N3-N7	25.000000

Resistor network v4 · Level 2 · task_0 · 18 resistors · 12 nodes

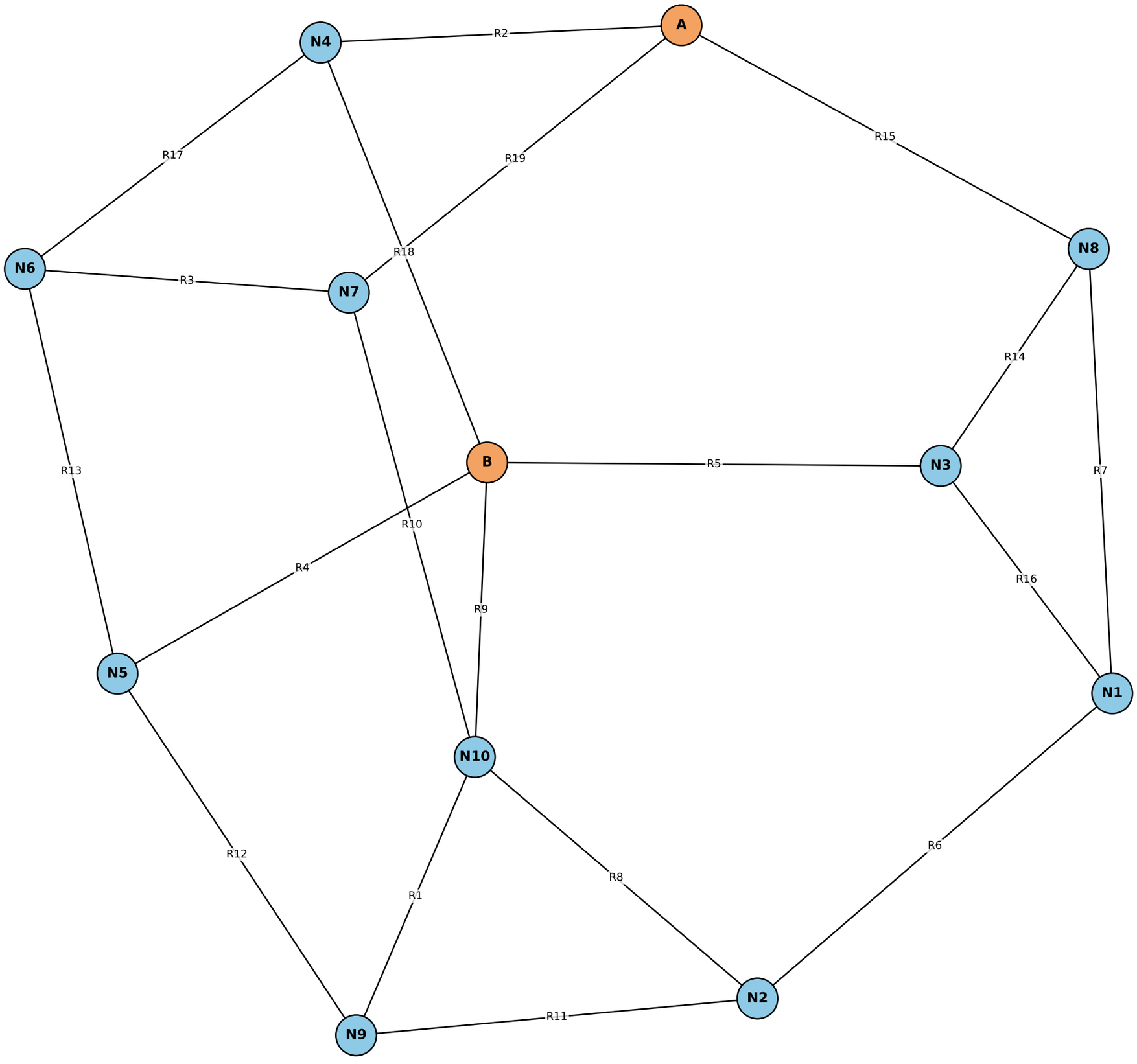
7 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	A-N10	35.000000
R2	N6-B	28.000000
R3	N2-N6	119.000000
R4	N3-B	180.000000
R5	N2-N7	11.000000
R6	N9-N10	31.000000
R7	N8-A	12.000000
R8	N5-N1	147.000000
R9	N5-N4	147.000000
R10	N3-N4	119.000000
R11	A-N1	28.000000
R12	N3-N8	15.000000
R13	N6-N7	31.000000
R14	N4-N1	31.000000
R15	N8-N5	35.000000
R16	N2-N9	35.000000
R17	N7-N10	47.000000
R18	N9-B	21.000000

Resistor network v4 · Level 2 · task_1 · 19 resistors · 12 nodes

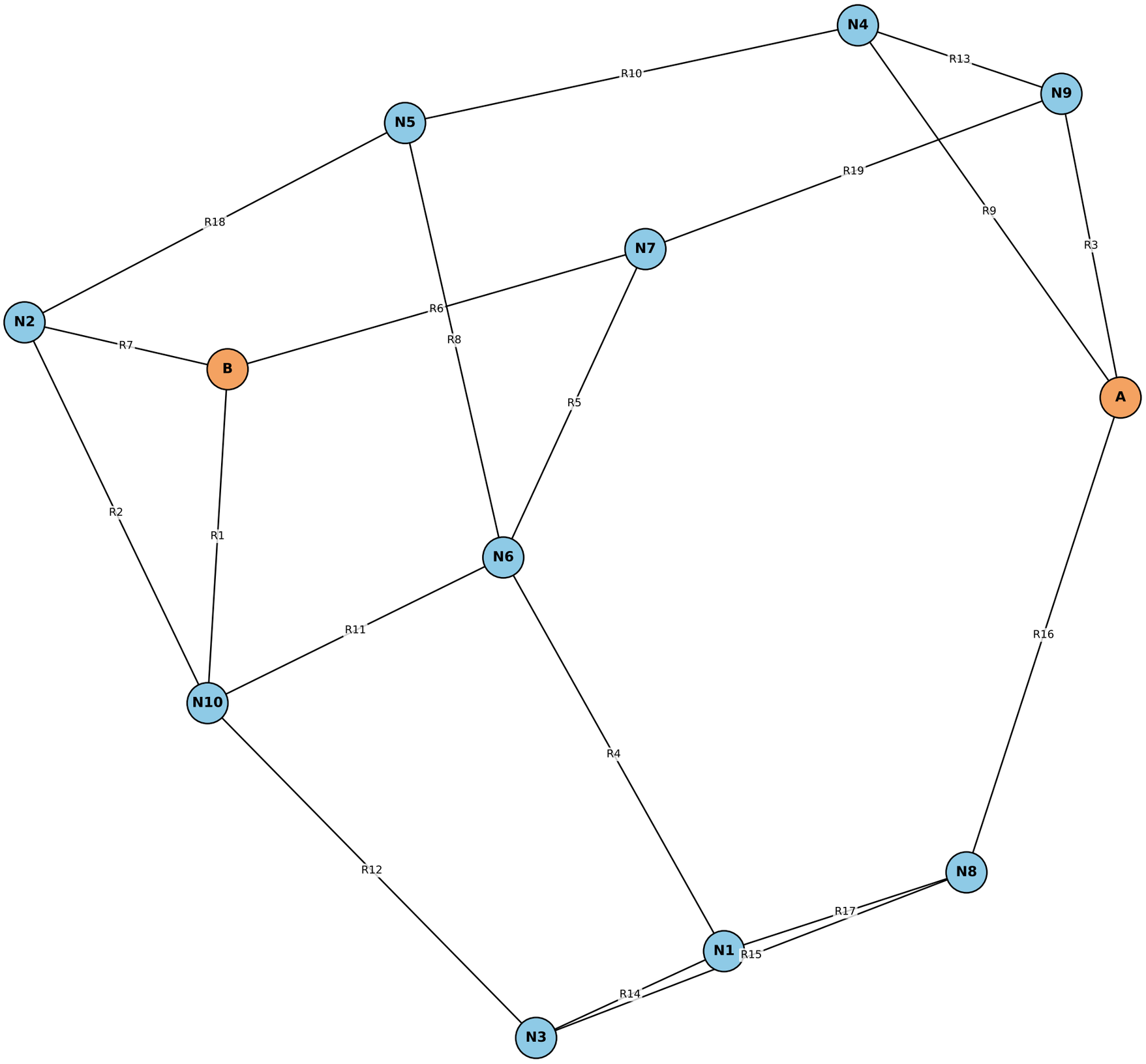
8 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N9-N10	88.000000
R2	N4-A	11.000000
R3	N6-N7	17.000000
R4	B-N5	14.000000
R5	B-N3	42.000000
R6	N2-N1	35.000000
R7	N8-N1	17.000000
R8	N2-N10	21.000000
R9	B-N10	180.000000
R10	N7-N10	108.000000
R11	N9-N2	28.000000
R12	N9-N5	119.000000
R13	N5-N6	147.000000
R14	N8-N3	21.000000
R15	N8-A	47.000000
R16	N1-N3	14.000000
R17	N4-N6	19.000000
R18	B-N4	47.000000
R19	A-N7	14.000000

Resistor network v4 · Level 2 · task_2 · 19 resistors · 12 nodes

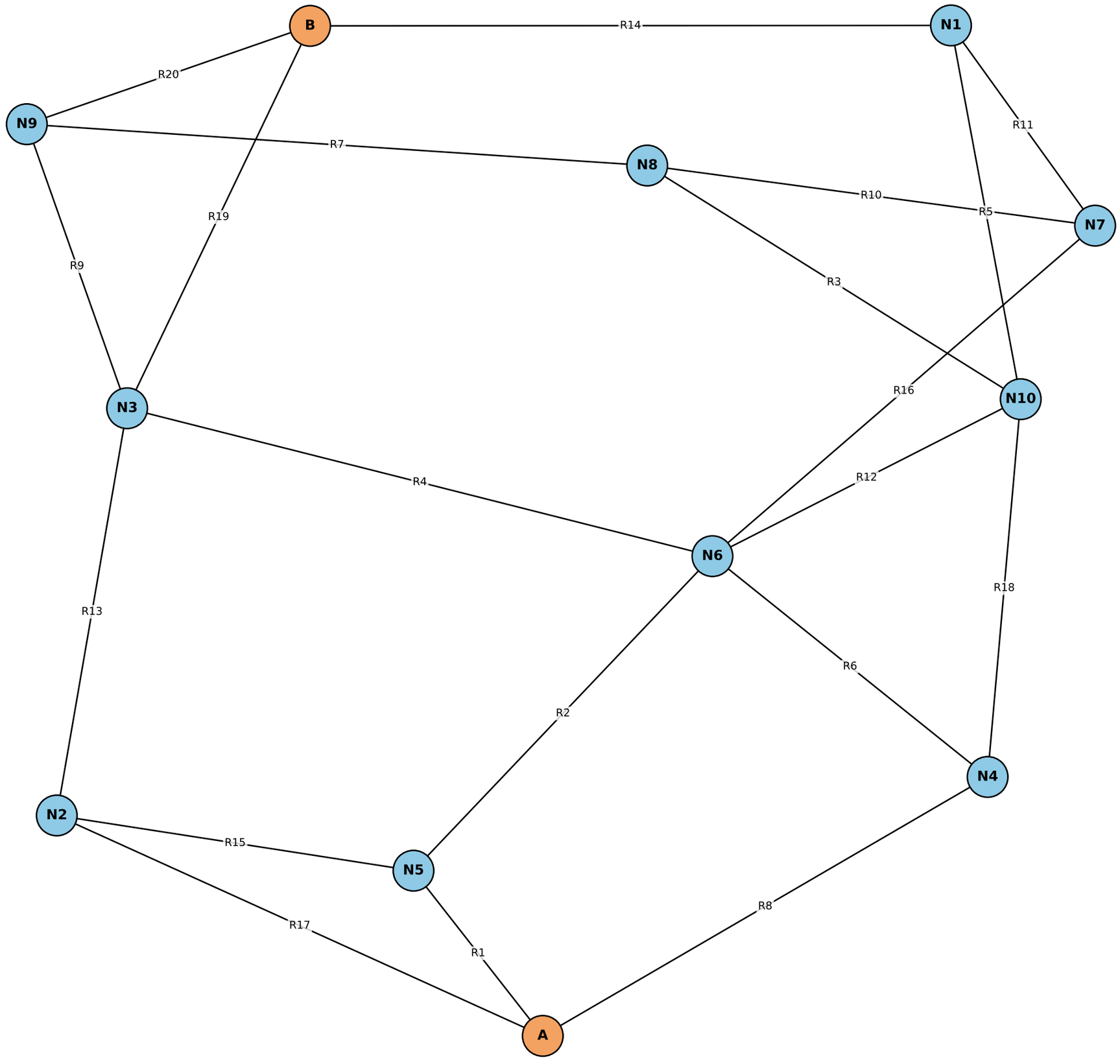
8 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N10-B	97.000000
R2	N10-N2	12.000000
R3	N9-A	38.000000
R4	N1-N6	58.000000
R5	N7-N6	38.000000
R6	N7-B	19.000000
R7	B-N2	25.000000
R8	N6-N5	15.000000
R9	N4-A	52.000000
R10	N4-N5	25.000000
R11	N6-N10	17.000000
R12	N10-N3	25.000000
R13	N9-N4	11.000000
R14	N1-N3	108.000000
R15	N8-N3	52.000000
R16	N8-A	21.000000
R17	N1-N8	42.000000
R18	N5-N2	200.000000
R19	N7-N9	28.000000

Resistor network v4 · Level 2 · task_3 · 20 resistors · 12 nodes

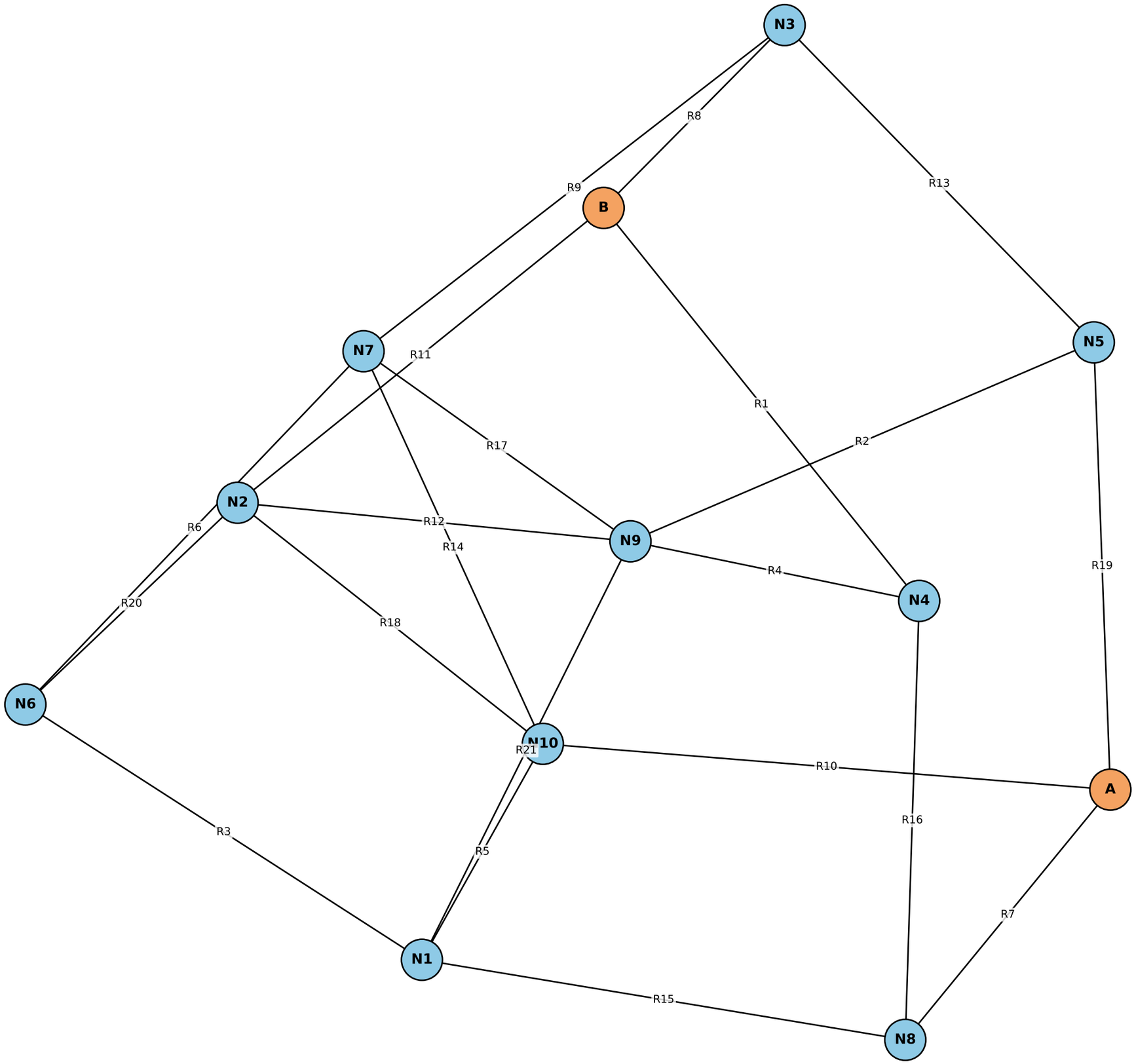
9 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	A-N5	119.000000
R2	N6-N5	147.000000
R3	N10-N8	35.000000
R4	N6-N3	88.000000
R5	N10-N1	10.000000
R6	N6-N4	19.000000
R7	N8-N9	19.000000
R8	N4-A	19.000000
R9	N3-N9	52.000000
R10	N8-N7	15.000000
R11	N7-N1	42.000000
R12	N6-N10	88.000000
R13	N2-N3	10.000000
R14	B-N1	79.000000
R15	N2-N5	108.000000
R16	N6-N7	25.000000
R17	N2-A	47.000000
R18	N10-N4	119.000000
R19	B-N3	88.000000
R20	B-N9	11.000000

Resistor network v4 · Level 2 · task_4 · 21 resistors · 12 nodes

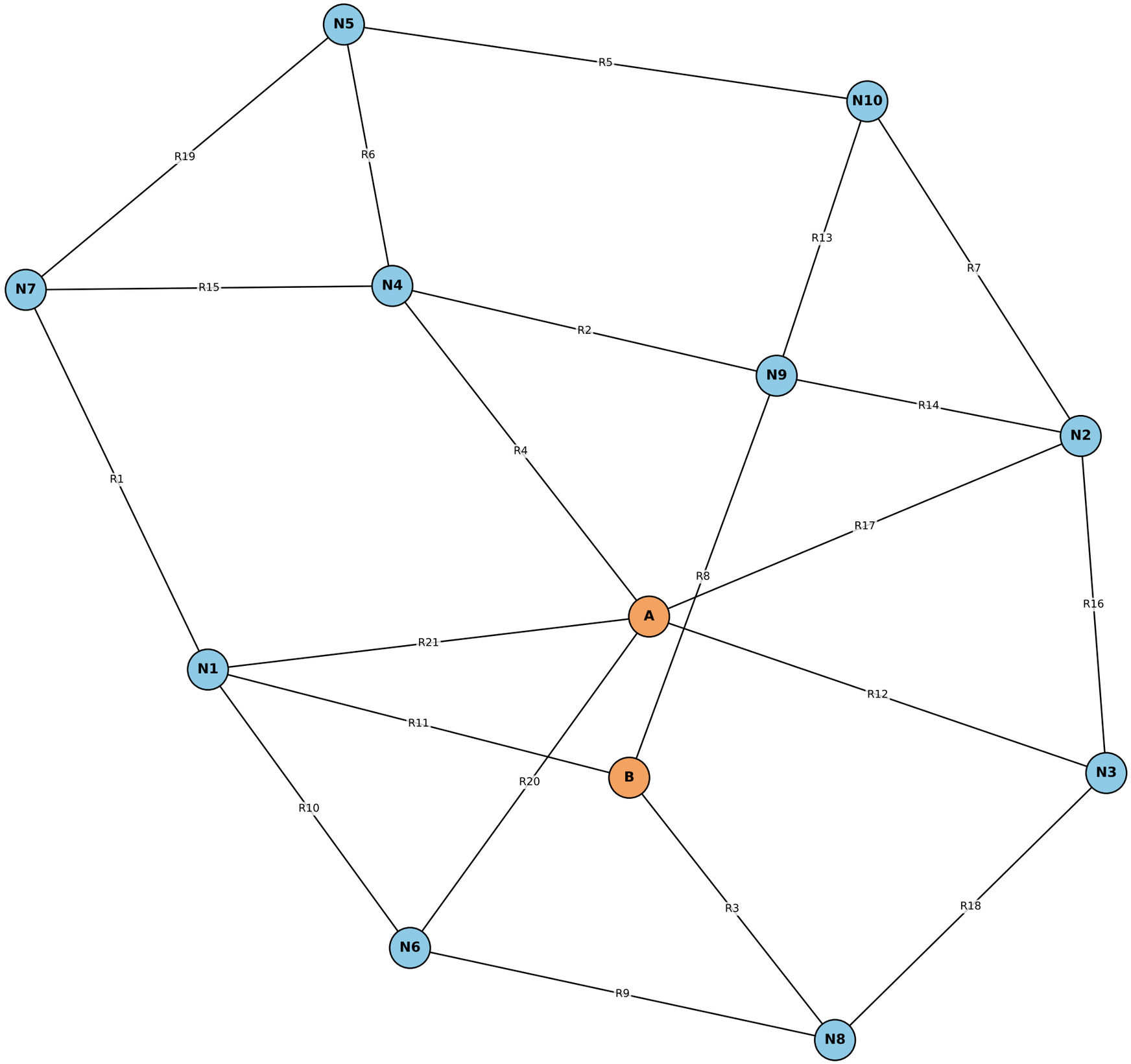
10 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N4-B	88.000000
R2	N9-N5	15.000000
R3	N1-N6	12.000000
R4	N9-N4	35.000000
R5	N10-N1	79.000000
R6	N7-N6	10.000000
R7	N8-A	47.000000
R8	N3-B	28.000000
R9	N7-N3	200.000000
R10	N10-A	163.000000
R11	B-N2	25.000000
R12	N9-N2	10.000000
R13	N5-N3	10.000000
R14	N10-N7	52.000000
R15	N8-N1	23.000000
R16	N8-N4	79.000000
R17	N9-N7	119.000000
R18	N10-N2	28.000000
R19	A-N5	97.000000
R20	N6-N2	79.000000
R21	N9-N1	71.000000

Resistor network v4 · Level 2 · task_5 · 21 resistors · 12 nodes

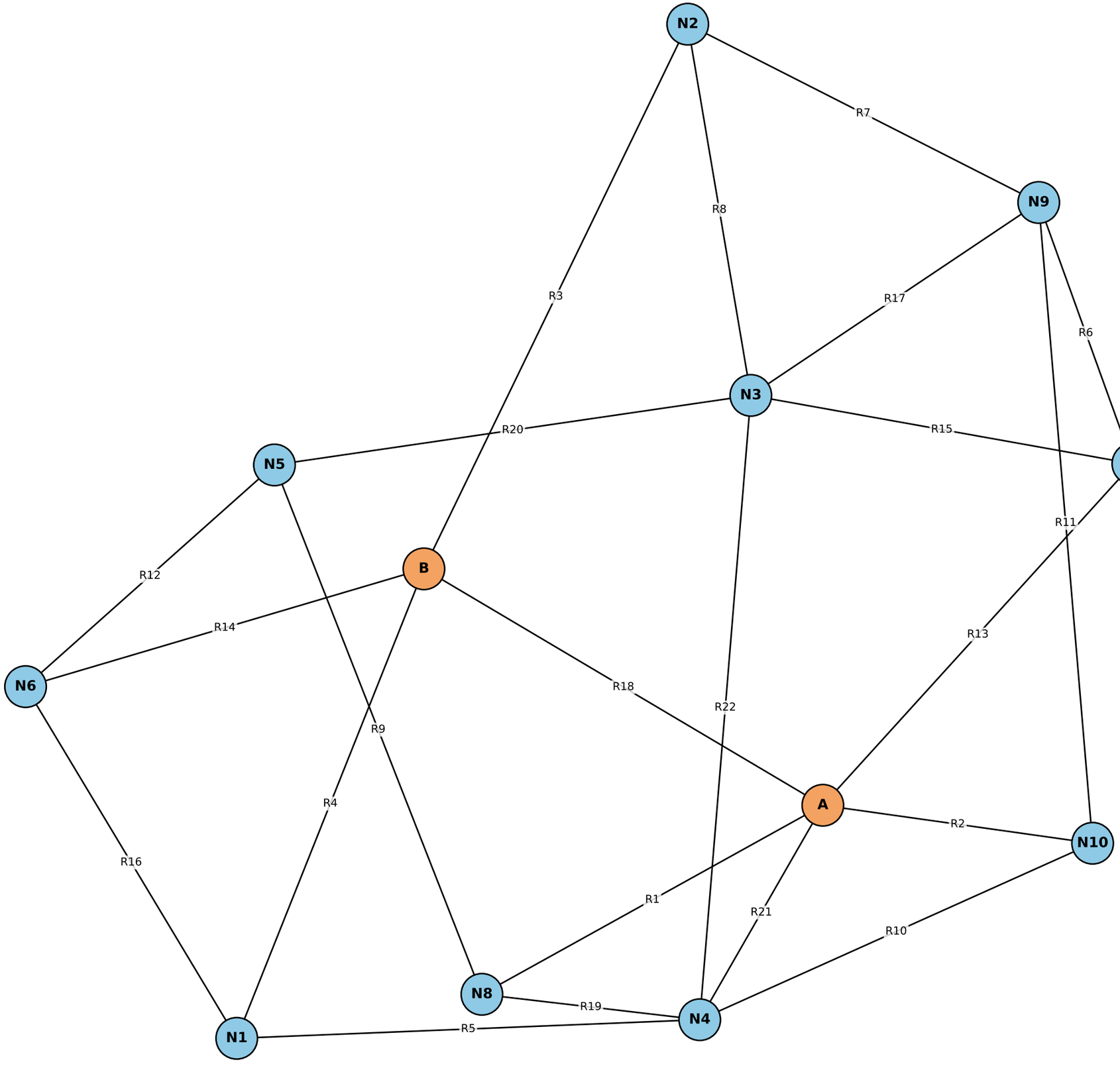
10 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N1-N7	10.000000
R2	N9-N4	31.000000
R3	N8-B	10.000000
R4	N4-A	19.000000
R5	N5-N10	97.000000
R6	N5-N4	147.000000
R7	N10-N2	200.000000
R8	B-N9	17.000000
R9	N8-N6	15.000000
R10	N1-N6	180.000000
R11	N1-B	79.000000
R12	N3-A	58.000000
R13	N10-N9	132.000000
R14	N9-N2	23.000000
R15	N7-N4	97.000000
R16	N3-N2	64.000000
R17	A-N2	14.000000
R18	N3-N8	28.000000
R19	N5-N7	52.000000
R20	N6-A	19.000000
R21	N1-A	147.000000

Resistor network v4 · Level 2 · task_6 · 22 resistors · 12 nodes

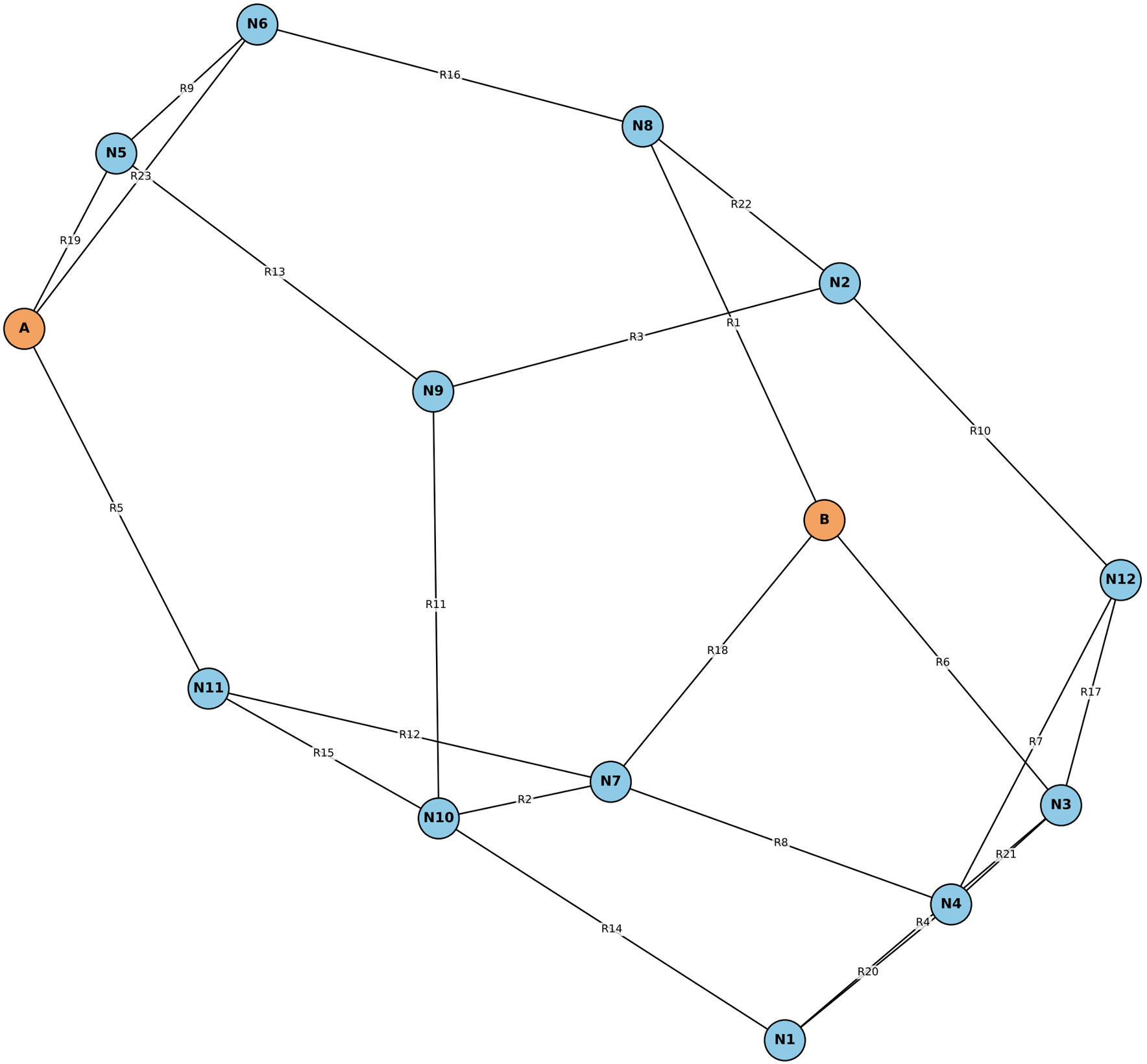
11 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	A-N8	108.000000
R2	A-N10	163.000000
R3	B-N2	17.000000
R4	B-N1	12.000000
R5	N1-N4	108.000000
R6	N7-N9	47.000000
R7	N2-N9	180.000000
R8	N3-N2	88.000000
R9	N8-N5	47.000000
R10	N4-N10	21.000000
R11	N9-N10	119.000000
R12	N6-N5	10.000000
R13	A-N7	52.000000
R14	B-N6	17.000000
R15	N3-N7	163.000000
R16	N6-N1	47.000000
R17	N3-N9	108.000000
R18	B-A	180.000000
R19	N8-N4	88.000000
R20	N3-N5	38.000000
R21	A-N4	35.000000
R22	N3-N4	79.000000

Resistor network v4 · Level 2 · task_7 · 23 resistors · 14 nodes

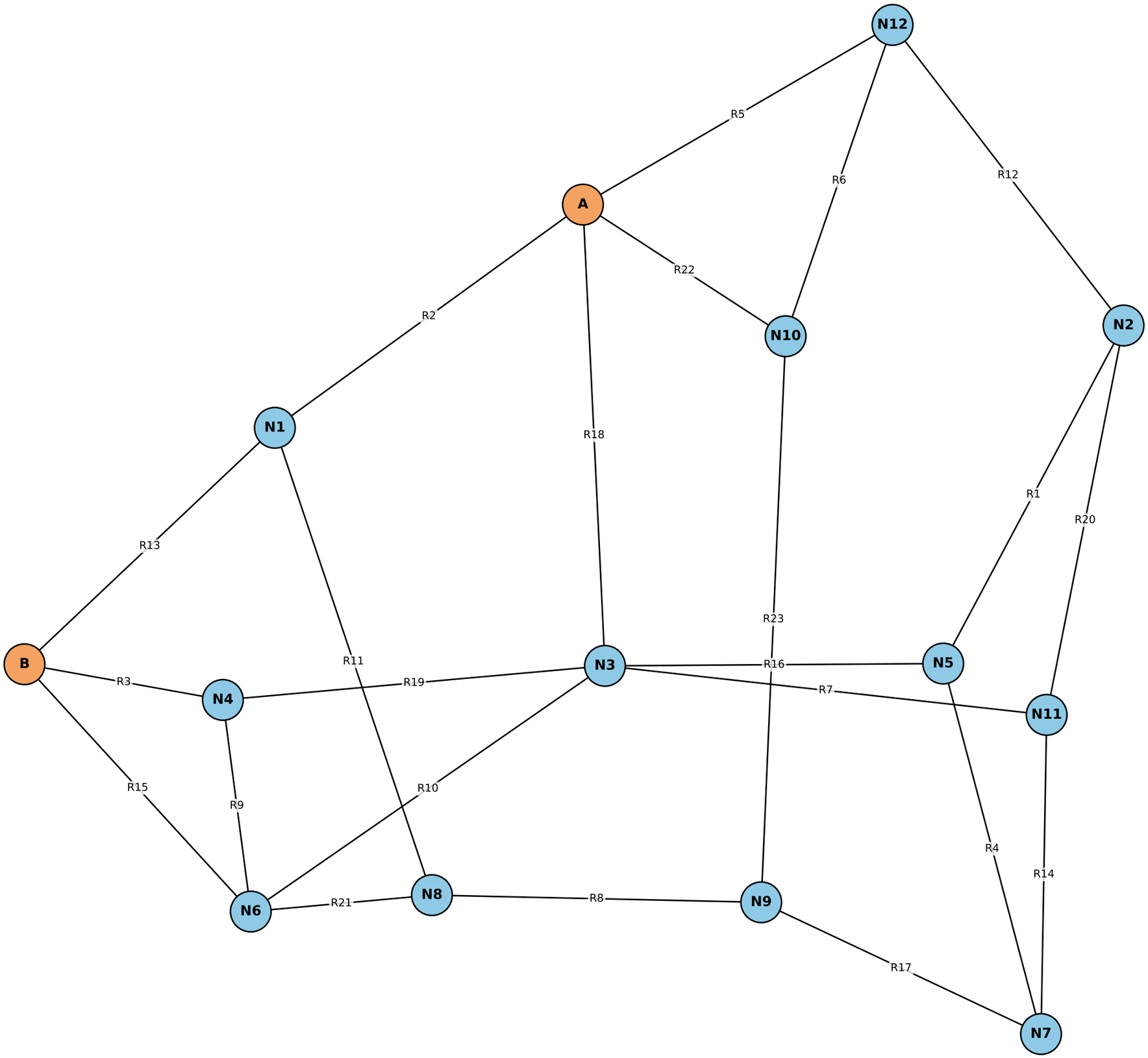
10 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	B-N8	200.000000
R2	N10-N7	19.000000
R3	N2-N9	119.000000
R4	N1-N3	12.000000
R5	N11-A	88.000000
R6	N3-B	147.000000
R7	N12-N4	180.000000
R8	N7-N4	23.000000
R9	N5-N6	132.000000
R10	N12-N2	12.000000
R11	N10-N9	42.000000
R12	N11-N7	31.000000
R13	N5-N9	19.000000
R14	N10-N1	119.000000
R15	N11-N10	28.000000
R16	N6-N8	28.000000
R17	N12-N3	21.000000
R18	N7-B	163.000000
R19	N5-A	47.000000
R20	N1-N4	11.000000
R21	N3-N4	38.000000
R22	N2-N8	132.000000
R23	N6-A	147.000000

Resistor network v4 · Level 2 · task_8 · 23 resistors · 14 nodes

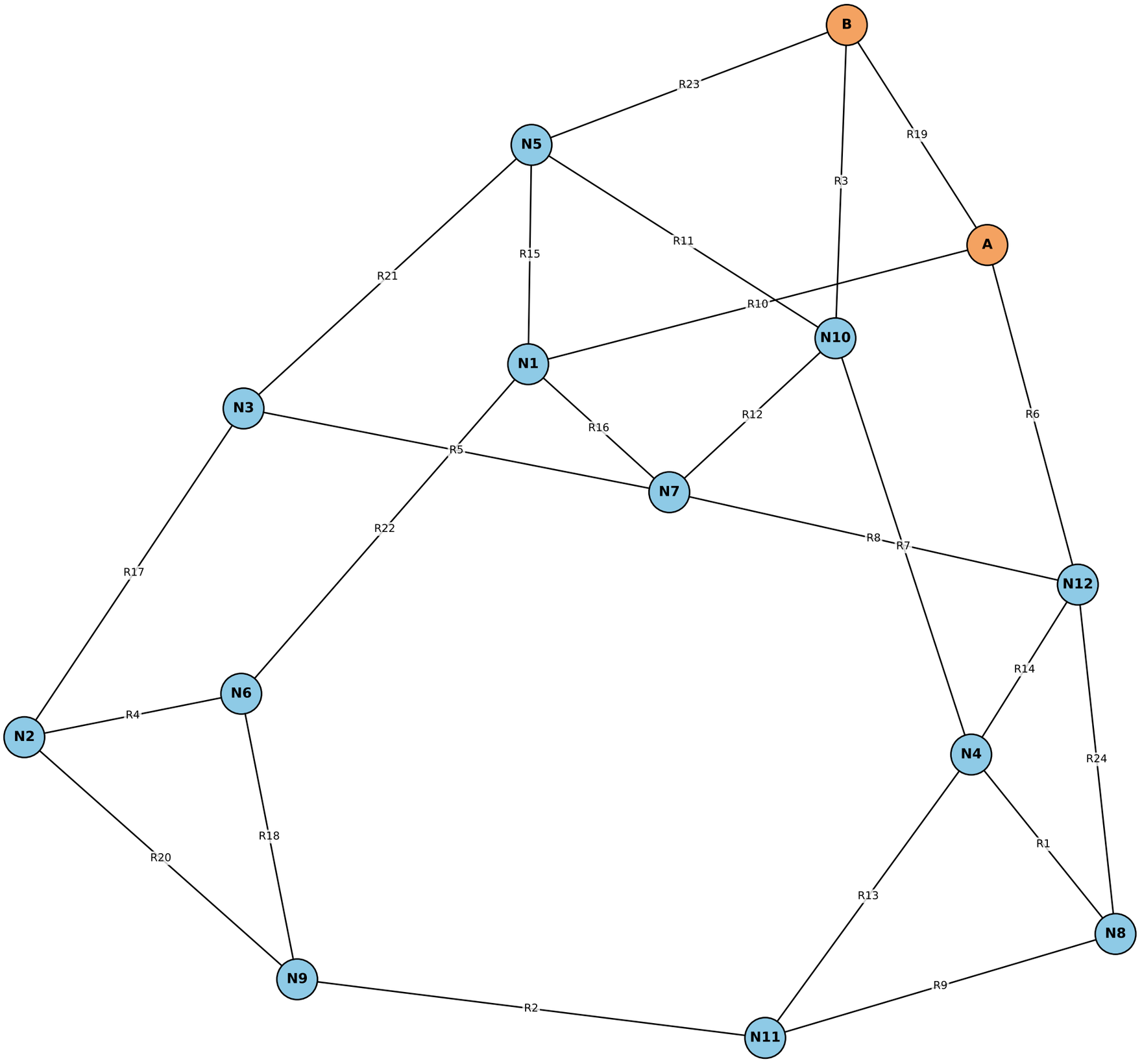
10 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N2-N5	21.000000
R2	A-N1	12.000000
R3	N4-B	19.000000
R4	N7-N5	28.000000
R5	N12-A	79.000000
R6	N10-N12	132.000000
R7	N3-N11	23.000000
R8	N9-N8	28.000000
R9	N4-N6	147.000000
R10	N3-N6	108.000000
R11	N8-N1	132.000000
R12	N12-N2	19.000000
R13	N1-B	52.000000
R14	N11-N7	52.000000
R15	B-N6	19.000000
R16	N3-N5	23.000000
R17	N7-N9	25.000000
R18	N3-A	25.000000
R19	N3-N4	14.000000
R20	N11-N2	108.000000
R21	N8-N6	147.000000
R22	N10-A	132.000000
R23	N10-N9	52.000000

Resistor network v4 · Level 2 · task_9 · 24 resistors · 14 nodes

11 independent cycles · A/B terminals in orange



ID	Nodes	Ohms
R1	N4-N8	42.000000
R2	N9-N11	88.000000
R3	N10-B	58.000000
R4	N6-N2	42.000000
R5	N3-N7	21.000000
R6	N12-A	79.000000
R7	N10-N4	47.000000
R8	N12-N7	19.000000
R9	N8-N11	58.000000
R10	N1-A	28.000000
R11	N10-N5	21.000000
R12	N10-N7	25.000000
R13	N4-N11	12.000000
R14	N12-N4	79.000000
R15	N1-N5	64.000000
R16	N1-N7	163.000000
R17	N3-N2	10.000000
R18	N6-N9	64.000000
R19	B-A	71.000000
R20	N2-N9	180.000000
R21	N3-N5	25.000000
R22	N6-N1	119.000000
R23	B-N5	35.000000
R24	N12-N8	47.000000